Morphological Processing

Week 7 — Shaping binary images

Practical Image Processing

Lecture 7 ▪ 60–70 min

Recap & today

- Wk2/8: thresholding gives a **binary** image — but it's messy (specks, gaps).
- Today: **morphology** — operations that grow, shrink, and clean binary shapes using a **structuring element**.

DocScan milestone

After thresholding the digits, morphology removes noise and fills gaps so each digit is a clean, separable blob.

Roadmap

The two basic operations

Opening & closing

More operations

In-class exercise

Deep dive

Wrap-up

The two basic operations

The structuring element

- A small shape (the **SE** / kernel) that slides over the binary image — like a probe.
- Common shapes: rectangle, ellipse, cross. Its **size** sets how aggressive the effect is.
- Morphology asks: does the SE *fit* (erosion) or *hit* (dilation) the shape here?

Math

`cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3,3))` — a 3×3 elliptical probe.

Erosion & dilation — on real digits



binary digits



erosion (thinner)



dilation (thicker)

- **Erosion**: keep a pixel only if the SE fits entirely inside the shape → shrinks, removes thin bits.
- **Dilation**: set a pixel if the SE touches the shape → grows, fills small holes.

Erosion and dilation are duals

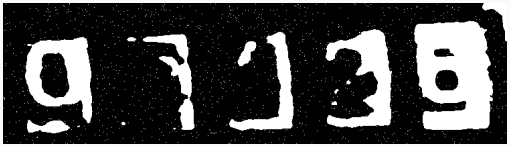
- Dilating the foreground = eroding the background (and vice versa).
- Erosion removes: thin protrusions, small objects, and it **separates** touching objects.
- Dilation adds: fills gaps, connects nearby pieces, and **thickens** thin strokes.

✕ Pitfall

Each also distorts: erosion can break a thin digit; dilation can merge two close digits. Size matters.

Opening & closing

Opening = erode then dilate



binary + speckles



opening: specks gone

- **Opening** removes small bright specks while keeping the digits' size (erode kills them, dilate restores the rest).
- The classic "clean up salt noise on a binary mask" operation.

Closing = dilate then erode



binary



closing: gaps filled

- **Closing** fills small holes and gaps inside shapes while keeping the size.
- Repairs a digit broken by thresholding into a solid stroke.

Opening vs closing: remember the effect

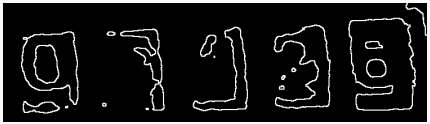
Operation	Order	Effect
Opening	erode $\rightarrow$ dilate	removes small *bright* specks, breaks thin links
Closing	dilate $\rightarrow$ erode	fills small *dark* holes, bridges gaps

Key idea

Opening cleans the outside; closing repairs the inside. Both keep the overall size, unlike raw erosion/dilation.

More operations

Morphological gradient, top-hat, black-hat



gradient = outline

- **Gradient** = dilation − erosion → the object outline.
- **Top-hat** = image − opening → small bright details (uneven lighting correction!).
- **Black-hat** = closing − image → small dark details.

✔ Good practice

Top-hat is a lighting fix: it isolates the digits from a slowly-varying background.

Grayscale morphology (brief)

- Morphology also works on grayscale: erosion = local min, dilation = local max.
- Top-hat on grayscale removes a smooth background gradient — great before thresholding.
- So morphology isn't only for binary — it's a general shape/intensity tool.

Structuring-element shapes

Shape	When	Note
Rectangle	general cleanup	fast, blocky effect
Ellipse	round objects	smoother, more natural
Cross	thin connectors	minimal, gentle
Line (h/v)	directional	remove/keep horizontal or vertical strokes

Key idea

A line SE can remove table lines while keeping text — the shape of the probe is a design choice.

Boundary extraction & hole filling

- **Boundary** = image − erosion (a thin outline; cf. the morphological gradient).
- **Hole filling**: fill enclosed dark regions inside a shape (a solid digit from a hollow one).
- These build on erosion/dilation — morphology is a small toolkit that composes into many effects.

In-class exercise

In-class exercise (12 min)

Pick the operation

Your thresholded meter has these problems. Which morphology fixes each?

1. Scattered single white pixels (salt) around the digits.
2. One digit is broken into two pieces by a threshold gap.
3. Two adjacent digits are touching and need to be separated for counting.
4. You want just the outline of each digit.

Debrief

- (1) Salt specks → **opening** (small SE). (2) Broken digit → **closing**.
- (3) Touching digits → **erosion** (carefully) to separate, or a watershed (Week 8).
- (4) Outline → **morphological gradient**.

Key idea

Diagnose the defect (specks / gaps / merges), then choose the matching operation and SE size.

Deep dive

Structuring element: shape & size matter

- A larger SE = a stronger effect (removes bigger specks / fills bigger gaps) but more distortion.
- Shape matters: a vertical SE affects vertical strokes differently than a horizontal one.
- Rule: use the *smallest* SE that fixes the defect — bigger erases real detail.

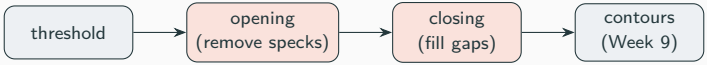
✕ Pitfall

A too-large SE turns your digits into unrecognizable blobs. Tune the size on real data.

Skeletonization & thinning

- **Skeleton**: repeatedly thin a shape to a 1-pixel "spine" — its topological essence.
- Useful for character/stroke analysis and measuring connectivity.
- Related: **hit-or-miss** finds exact local patterns (corners, endpoints).

A morphology cleanup pipeline



- Typical DocScan order: threshold → open (kill noise) → close (repair digits) → contours.
- Order matters: opening before closing avoids fusing a speck into a digit.

Removing lines, keeping text

- A form has table/underline strokes you want gone but the text kept.
- Use a long *horizontal* line SE with opening to isolate horizontal lines, then subtract them.
- A directional SE is a precise tool — it operates only on structures of a chosen orientation and length.

✓ Good practice

For a scanned form (a DocScan case), this cleanly separates rules/boxes from the handwritten content.

Morphology vs blurring

- Blurring averages (softens everything); morphology reasons about *shape* (keeps edges hard).
- To remove specks from a *binary* mask, opening beats a blur (which would gray it out).
- Use blur on grayscale/noise; use morphology on binary shapes.

Key idea

Morphology is the right tool once you have a binary mask — it's shape surgery, not smoothing.

Wrap-up

What you should now be able to do

- Explain erosion/dilation via a structuring element (fit vs hit).
- Use opening (remove specks) and closing (fill gaps); pick the SE size.
- Apply gradient/top-hat/black-hat; correct uneven lighting with top-hat.
- Build a threshold $\rightarrow$ morphology cleanup pipeline for the digits.

Morphology in code

```python
k = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (3, 3))
op  = cv2.morphologyEx(binary, cv2.MORPH_OPEN,  k)   # remove specks
cl  = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, k)   # fill gaps
top = cv2.morphologyEx(gray,   cv2.MORPH_TOPHAT, big_k)  # lighting fix
```

`morphologyEx` covers open/close/gradient/top-hat/black-hat with one call.

This week's deliverable

Due before Week 8 (see `project/`)

On a thresholded meter/plate (from Week 2):

1. Demonstrate erosion, dilation, opening, closing; explain each effect.
2. Use opening to remove speckle noise and closing to repair broken digits.
3. Apply top-hat to a photo with uneven lighting; compare to before.
4. DocScan milestone: produce a clean binary where each digit is a solid, separable blob.

Threshold makes it binary; morphology makes it clean.

Opening removes specks, closing fills gaps — shape surgery.